The Chain Rule & Directional Derivatives

Section 1: Multivariable Chain Rule

1 If $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$, what is $\frac{dz}{dt}$?

a. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$

b. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dy}{dt} + \frac{\partial z}{\partial y}\frac{dx}{dt}$

c. $\frac{dz}{dt} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\frac{dz}{dt} = \frac{dx}{dt}\frac{dy}{dt}$

2 Given $z = x^2y$, $x = t^3$, and $y = t^2$, find $\frac{dz}{dt}$ at $t = 1$.

a. 8

b. 6

c. 5

d. 12

3 If $z = f(x, y)$ with $x = r\cos\theta$ and $y = r\sin\theta$, what is $\frac{\partial z}{\partial r}$?

a. $\frac{\partial z}{\partial x}\cos\theta + \frac{\partial z}{\partial y}\sin\theta$

b. $-r\frac{\partial z}{\partial x}\sin\theta + r\frac{\partial z}{\partial y}\cos\theta$

c. $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\cos\theta + \sin\theta$

Section 2: The Gradient Vector

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4 What is the gradient vector $\nabla f(x, y)$ of a scalar function $f(x, y)$?

a. $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$

b. $\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

c. $\nabla f = \left[\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}\right]$

d. $\nabla f = \sqrt{f_x^2 + f_y^2}$

5 Compute the gradient $\nabla f(2, -1)$ for $f(x, y) = 3x^2y + 2y^2$.

a. $[-12, 8]$

b. $[12, -8]$

c. $[-12, -4]$

d. $[6, 4]$

6 What is the geometric relationship between the gradient $\nabla f(x_0, y_0)$ and the level curve $f(x, y) = c$ passing through (x_0, y_0)?

a. $\nabla f(x_0, y_0)$ is perpendicular (orthogonal) to the tangent line of the level curve

b. $\nabla f(x_0, y_0)$ is parallel to the tangent line

c. $\nabla f(x_0, y_0)$ has zero dot product with the normal vector

d. $\nabla f(x_0, y_0)$ points along the curve

Section 3: Directional Derivatives

7 What is the formula for the directional derivative $D_{\mathbf{u}} f$ in the direction of a unit vector $\mathbf{u}$?

a. $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$

b. $D_{\mathbf{u}} f = \nabla f \times \mathbf{u}$

c. $D_{\mathbf{u}} f = \|\nabla f\| \|\mathbf{u}\|$

d. $D_{\mathbf{u}} f = \frac{\nabla f \cdot \mathbf{u}}{\|\nabla f\|}$

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8 If $\nabla f(1,2) = [4, -3]$ and $\mathbf{u} = \left[\frac{3}{5}, \frac{4}{5}\right]$, find $D_{\mathbf{u}} f(1,2)$.

a. 0

b. 5

c. $\frac{24}{5}$

d. -1

9 Why must $\mathbf{u}$ be normalized to a unit vector before computing $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$?

a. To ensure the derivative measures rate of change per unit of geometric distance

b. Because dot products cannot be computed with non-unit vectors

c. To avoid division by zero

d. Because the gradient has length 1

Section 4: Direction of Maximum Increase & Steepest Ascent

10 In which direction does a differentiable function f increase most rapidly at a point?

a. In the direction of the gradient vector ∇f

b. In the direction opposite to ∇f (i.e. $-\nabla f$)

c. Orthogonal to the gradient vector

d. Along the tangent to the level curve

11 What is the maximum value of the directional derivative $D_{\mathbf{u}} f$ at a point?

a. $\|\nabla f\|$

b. $\|\nabla f\|^2$

c. 0

d. 1

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12 In which direction is the rate of change of f equal to zero?

a. In any direction tangent to the level curve (orthogonal to ∇f)

b. In the direction of $-\nabla f$

c. In the direction of ∇f

d. Nowhere, a differentiable function always changes

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